



Antimicrobial Drugs: Mechanisms of Action

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Introduction

Antimicrobial drugs are chemical substances of **natural** or **synthetic origin** act by killing or interfering with the growth of microorganisms.

These are of a few types which basically include:

- Antibacterial
- Antifungal
- Antivirus
- Antihelmentic
- Antiprotozoal drugs



Antibiotics

- Type of antimicrobial substance active against bacteria.
- Widely used in the treatment and prevention of bacterial infections.
- Not effective against viruses such as the common cold or influenza.



- **Antibacterial** drugs constitute the **greatest number** and diversity of antimicrobial agents (Because there are many differences between the structure and metabolism of pathogenic bacteria and their eukaryotic hosts.
- Fewer **antifungal, antiprotozoan, and anthelmintic** drugs are available because fungi, protozoa, and helminthes like their animal and human hosts are eukaryotes.
- The number of effective **antiviral drugs** is also limited, despite major differences in structure, because viruses utilize their host cells' enzymes and ribosomes to metabolize and replicate. Therefore, drugs that are effective against viral replication are likely toxic to the host as well.

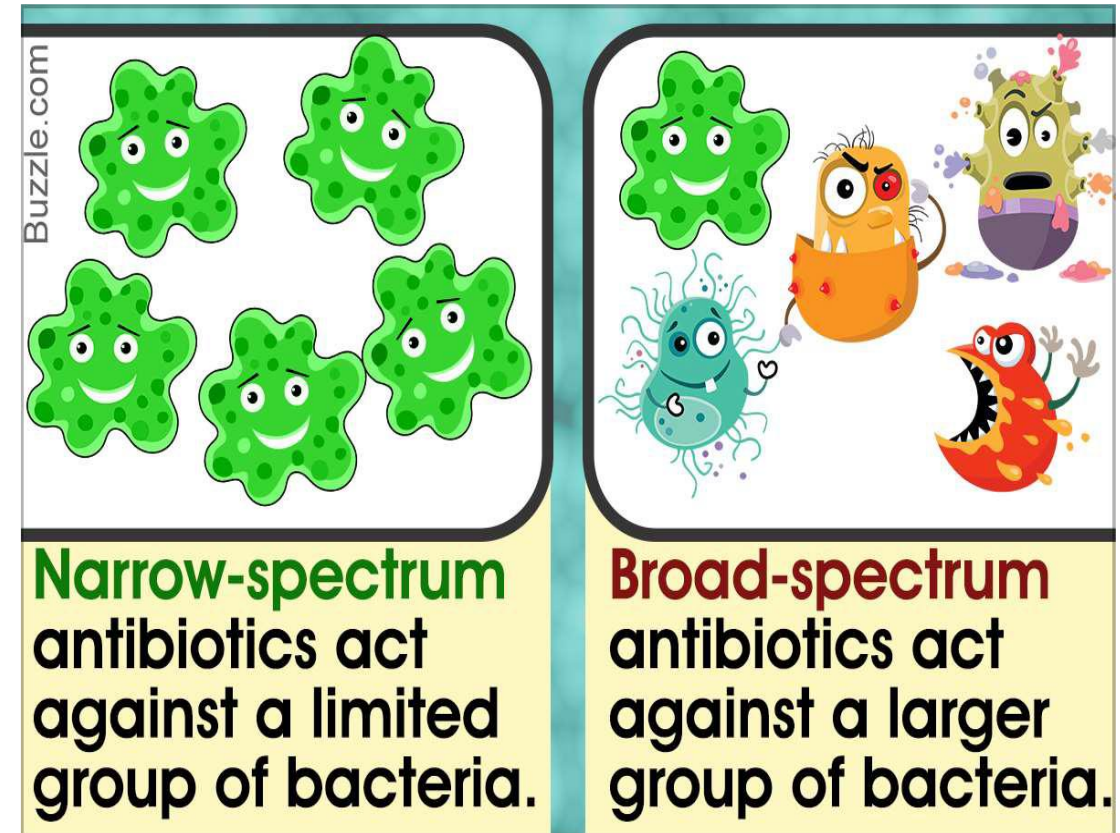
Principles of Antimicrobial therapy

The key to successful chemotherapy against microbes is **selective toxicity**

- **Selective toxicity:** selective inhibition of the growth of the microorganisms without damage to the host.
- There are four major sites in the bacterial cell that are sufficiently different from the human cell that they serve as the basis for the action of clinically effective drugs: **cell wall, ribosomes, nucleic acids, and cell membrane.**

Classification of Antibiotics

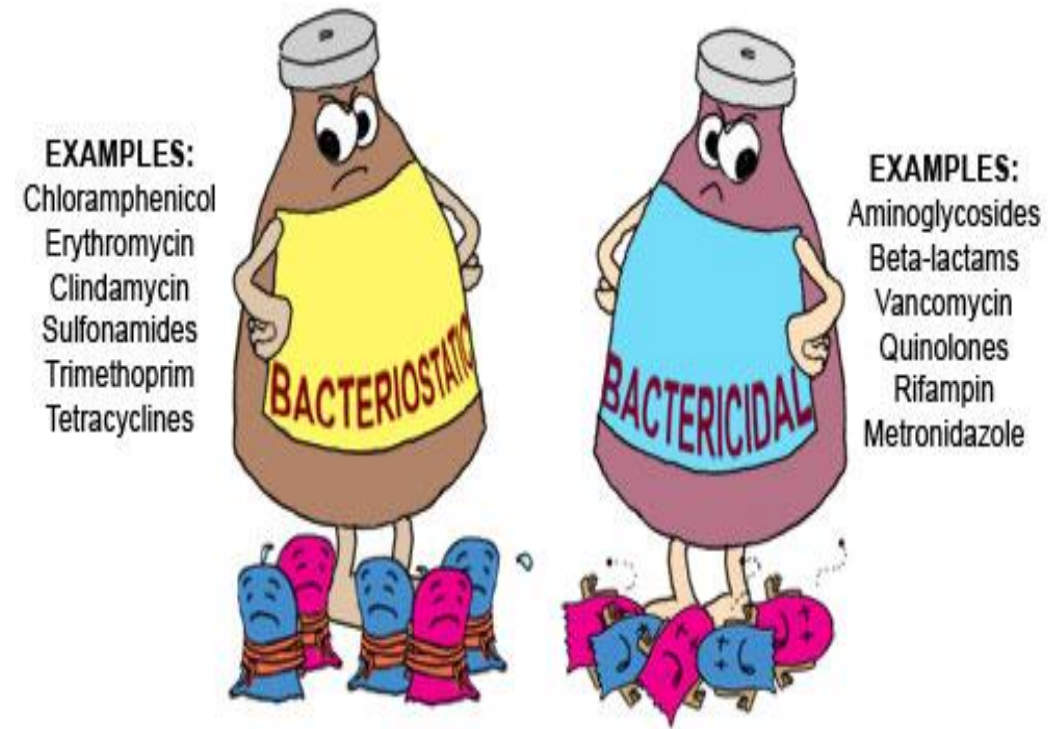
- **Based on their spectrum of action:** Range of activity of an antimicrobial against bacteria.
- **Broad-spectrum** antibiotics are active against several types of microorganisms.
- **Narrow-spectrum** antibiotics are active against one or very few types.



Classification of Antibiotics

➤ Based on mod of action

- **Bactericidal drugs** kill bacteria, whereas **bacteriostatic drugs** inhibit the growth of the bacteria but do not kill.
- **Bacteriostatic drugs** depend on **the phagocytes** of the patient to kill the organism. If a patient has **too few neutrophils**, then bactericidal drugs should be used.

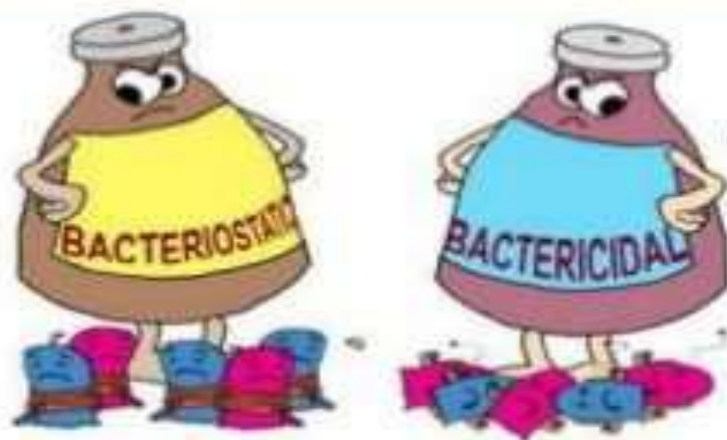


Classification of Antibiotics

Based on mode
of Action

Bacteriostatic

Bactericidal



Based on their
spectrum of
action

Broad-spectrum

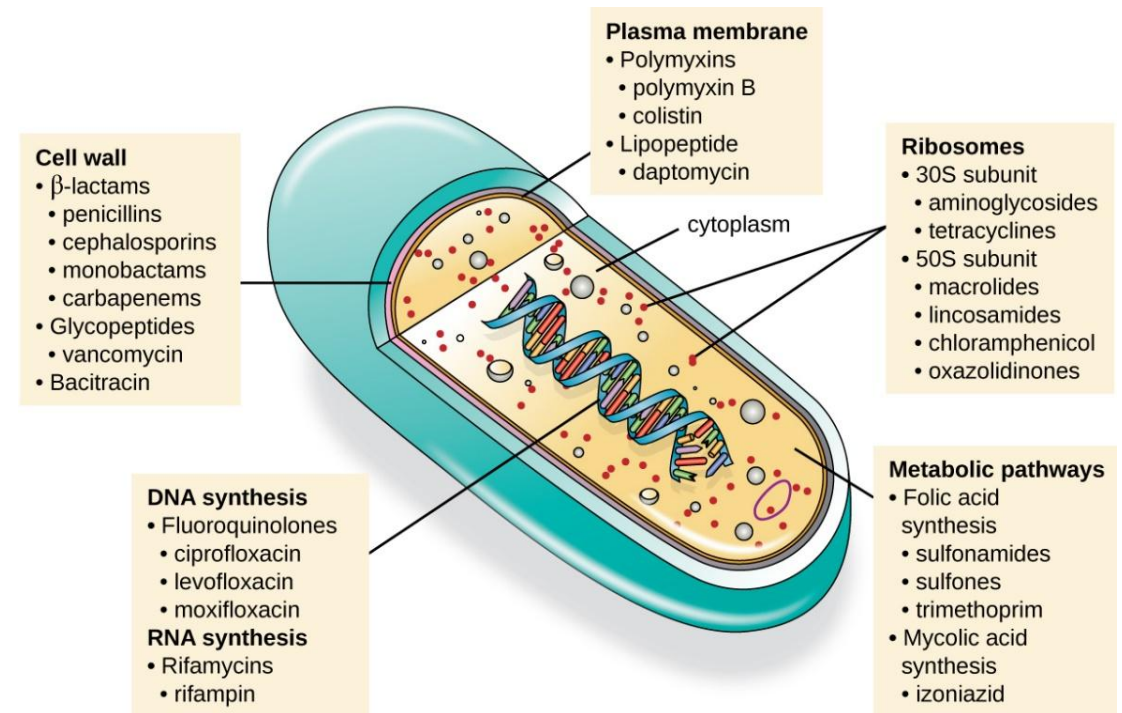
Narrow
Spectrum



Mechanisms of action

Antimicrobial drugs act in one of several ways: by selective toxicity, by

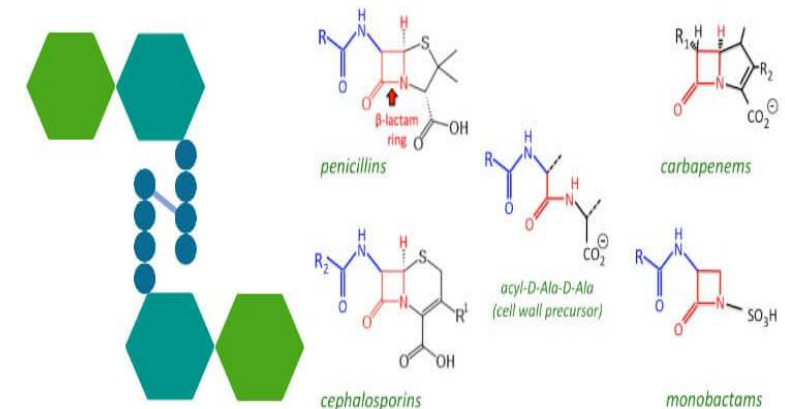
- Inhibition of Cell Wall Synthesis
- Inhibition of Protein Synthesis
- Inhibition of Nucleic Acid Synthesis
- Inhibition/Alteration of Cell Membrane Function



Mechanisms of action

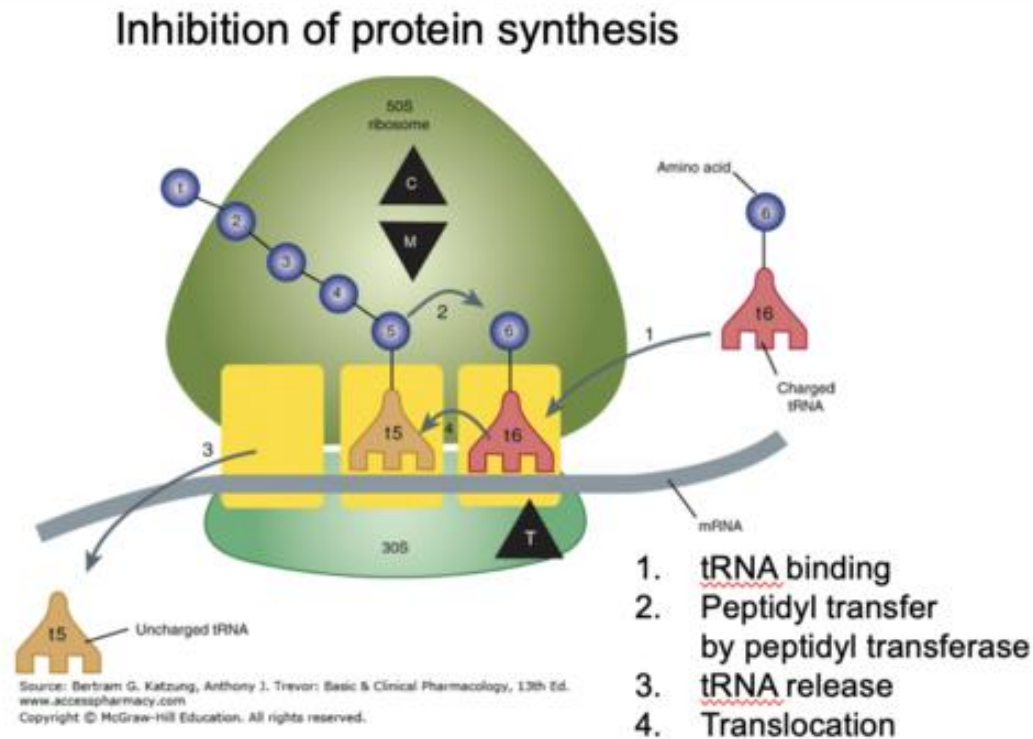
Inhibition of Cell Wall Synthesis

- Antimicrobial drugs act by inhibiting transpeptidases, the enzymes that catalyze the final cross-linking step in the synthesis of peptidoglycan.
- Example: **β -Lactam antibiotics**
 - Include penicillins, cephalosporins, and cephameycins
 - Over half of all antibiotics used worldwide



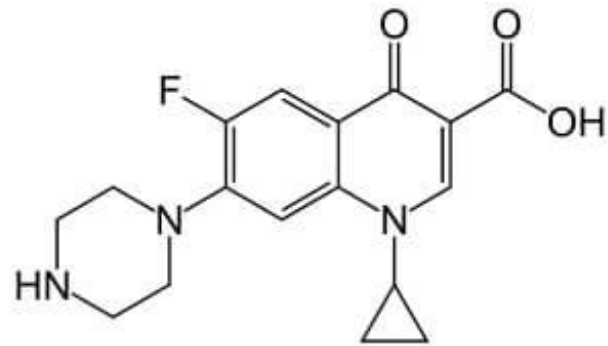
- Penicillin kills cells only **when they are growing**. When cells are growing, **new peptidoglycan** is being synthesized, and transpeptidation occurs. However, in non-growing cells, no new cross-linkages are required, and penicillin is inactive.
- Penicillins are usually **nontoxic** at clinically effective levels. The major disadvantage of these compounds is **hypersensitivity**, with a reported prevalence of 1% to 10% of patients.
- Exposure of the bacterial cell to **lysozyme**, which is present in **human tears**, results in **degradation of the peptidoglycan** and osmotic rupture similar to that caused by penicillin.

Inhibition of Protein Synthesis

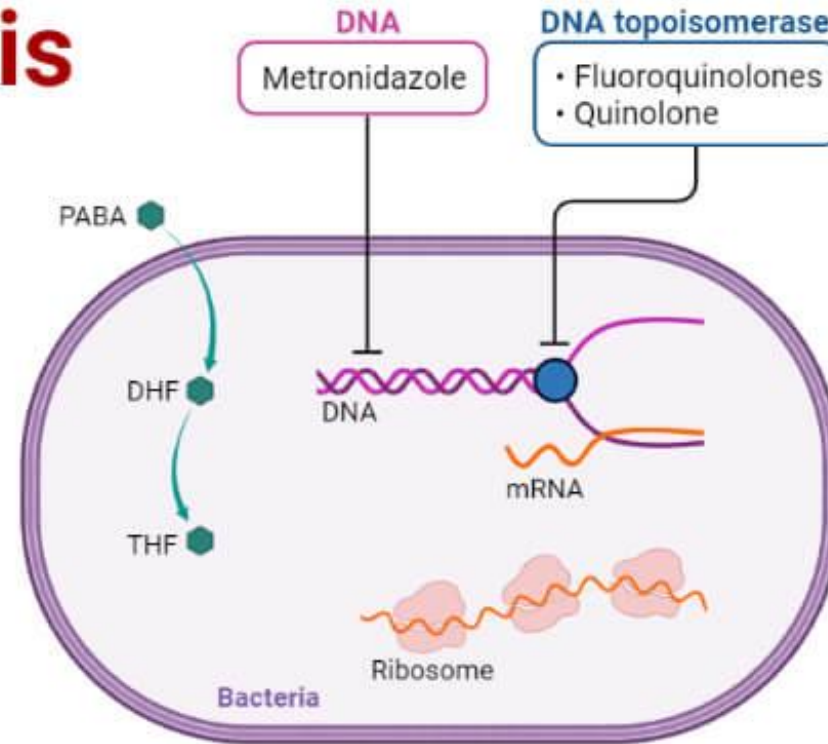


- **Aminoglycosides & tetracyclines** act at the level of the 30S ribosomal subunit.
- **chloramphenicol, erythromycins, and clindamycin** act at the level of the 50S ribosomal subunit.

DNA Synthesis Inhibitors



Ciprofloxacin



- Inhibition of Precursor Synthesis : **Sulfonamides & Trimethoprim**
- Inhibition of DNA Synthesis: **Fluoroquinolones** block bacterial DNA synthesis by inhibiting **DNA gyrase** such as ciprofloxacin
- Inhibition of mRNA Synthesis: **Rifampin**

Alteration of Cell Membrane Function

1- Alteration of Bacterial Cell Membranes: There are few antimicrobial compounds that act on the cell membrane because the structural and chemical similarities of bacterial and human cell membranes make it difficult to provide sufficient selective toxicity:

- **Polymyxins** (It is active against gram-negative rods).
- **Daptomycin** (It is active against gram-positive cocci).

2- Alteration of Fungal Cell Membranes: **Amphotericin B** (disrupts fungal cell membranes by binding at the site of ergosterol in the membrane).

Additional Drug Mechanisms

- **Isoniazid** inhibits the synthesis of **mycolic acid**; a long chain fatty acid found in the cell wall of mycobacteria. Isoniazid is the most important drug used in the treatment of **tuberculosis** and other mycobacterial diseases.
- **Metronidazole** is effective against anaerobic bacteria and certain protozoa because it **acts as an electron sink**, taking away the electrons that the organisms need to survive. It also forms toxic intermediates that damage DNA.

chemoprophylaxis

- ❑ Antimicrobial agents are used for the *treatment* of infectious diseases. However, there are times when they are used to *prevent* diseases from occurring—a process called **chemoprophylaxis**.
- ❑ Chemoprophylaxis is used in **three** situations:
 - Prior to surgery,
 - Immunocompromised patients,
 - People with normal immunity who have been exposed to certain pathogens.

Probiotics

- ❑ Live, nonpathogenic bacteria (or yeasts) that may be effective in the treatment or prevention of certain human diseases.
- ❑ The suggested basis for the possible beneficial effect lies either in
 - The nonpathogen excludes the pathogen from binding sites.
 - Reducing the inflammatory response against the pathogen.
- ❑ For example, the oral administration of live *Lactobacillus rhamnosus* strain GG significantly reduces the number of cases of nosocomial diarrhea in young children.

Q1/ The selective toxicity of antifungal drugs, such as amphotericin B and itraconazole, is based on the presence in fungi of which one of the following?

- A. 30S ribosomal subunit
- B. Dihydrofolate reductase
- C. DNA gyrase
- D. Ergosterol
- E. Mycolic acid

Q2/ Q4/ Which of the following pathogens is never treated with antibiotics?

- A. Streptococcus pneumoniae
- B. Mycobacterium tuberculosis
- C. Bordetella pertussis
- D. Respiratory syncytial virus
- E. Bacterial meningitis

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